

MOHAMMED BILAAL BASHA A

Mechatronics Engineer & Software Developer

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PROFESSIONAL SUMMARY

Driven Mechatronics Engineering student with hands-on experience across 5 internships in Embedded Systems, AI/NLP, Cloud Computing, and Industrial Design. Demonstrated success in leading projects from concept to deployment, including camera-based classroom automation and deep learning security systems. Combines technical expertise with strong problem-solving and communication skills to bridge hardware engineering and software solutions.

SKILLS & COMPETENCIES

Languages : Embedded C, C++, Python, Java, HTML
Engineering & Tools: SolidWorks (3D Modeling), MS Excel
Cloud & AI: AWS Fundamentals, Microsoft Azure AI, Deep Learning, NLP
Soft Skills: Communication, Critical Thinking, Problem Solving, Leadership

PROFESSIONAL INTERNSHIPS

- Embedded Systems Intern • CommonRocket** On-site | Jun 2026
- Gained hands-on experience with STM32 microcontrollers, Embedded C, STM32CubeIDE, GPIO, timers, PWM, UART, motor control, peripheral interfacing, and hardware debugging.
 - Developed practical knowledge of DC and stepper motor control using driver modules, applying embedded concepts to industrial automation and control applications through programming, testing, and debugging.
- AI Text Summarization Intern • AICTE** Online | Dec 2025
- Developed and analyzed AI-based text summarization models utilizing machine learning algorithms and Natural Language Processing (NLP) pipeline architectures.
 - Gained hands-on experience structuring Natural Language Processing (NLP) pipeline architectures and model.
- Industrial Design Intern • TITAN Watches** On-site | Jun 2025
- Observed industrial-scale product design pipelines and collaborated with design teams to optimize operational workflows.
 - Analyzed physical manufacturing layouts to identify, report, and help resolve operational bottlenecks.
- IT Project Intern • Dia Infra Solutions Pvt. Ltd.** On-site | Aug 2024
- Acquired key insights into conceptual design layouts and standard IT-driven project management methodologies.
 - Assisted in documenting structured project milestone plans and technical workflow blueprints.
- AWS Cloud Intern • Internship Studio** Online | Mar 2024
- Completed practical hands-on training on fundamental AWS operations and infrastructure deployment.
 - Built and tested virtualized computing environments, databases, and secure networking architectures using AWS core services

PROJECTS

- IoT & AI-Powered Intelligent Classroom Management System [Final Year Project]**
- Implemented camera-based student counting using AI/Computer Vision to estimate classroom occupancy and leveraged occupancy data to optimize energy usage of lights, fans, and AC.
- AI-Powered Digital Identity Protection Application [Deep Learning]**
- Engineered an AI application using deep learning facial recognition models to detect unauthorized image use across digital platforms and send security alerts.
- Line Following Robot [Embedded Systems / IoT]**
- Constructed an autonomous mobile robot using IR sensor feedback and programmed embedded control logic for real-time speed control and directional path correction.
- Personal Expense Tracker Web App [Full-stack Web Dev]**
- Developed a responsive web-based tracking dashboard for personal budget management using HTML and robust transaction-handling algorithms in Java.

EDUCATION

B.E. Mechatronics Engineering
SNS College of Technology, Coimbatore
CGPA: Above 8.40/10.0 (till Sem VI) | 2023 – 2027

HSC — Green Valley Mat. Hr. Sec School
Score: 80% | 2021 – 2023

SSLC — St. Joseph Mat. Hr. Sec School
Score: 80% | 2020 – 2021

CERTIFICATIONS & CO-CURRICULARS

Microsoft Azure AI Fundamentals (AI-900)
Credential ID: Lx6y-s4wW | Oct 2025

SolidWorks 3D Modeling
(3 - Day intensive Training | sep 2025)

Co-Curricular:
ROTA QUEST Quizzathon | Jan 2024

LANGUAGE
English, Hindi, Tamil, Urdu

DECLARATION

I hereby declare that the information provided above is true to the best of my knowledge and belief.